Prob 1 - Polynomial interpolation

$$F(x) = 3^{-(x-1)^2}$$

	x_0	x_1	x_2	x_3
x	-1	0	1	2
y	$1/81$	$1/3$	1	$1/3$

$$l_0 = \left(\frac{x-x_1}{x_0-x_1}\right)\left(\frac{x-x_2}{x_0-x_2}\right)\left(\frac{x-x_3}{x_0-x_3}\right) = \overset{-x}{\left(\frac{x-0}{-1-0}\right)}\overset{-\frac{x-1}{2}}{\left(\frac{x-1}{-1-1}\right)}\overset{-\frac{x-2}{3}}{\left(\frac{x-2}{-1-2}\right)} = -\frac{1}{6}\left(x(x-1)(x-2)\right) = -\frac{x^3}{6} + \frac{x^2}{2} - \frac{x}{3}$$

$$l_1 = \left(\frac{x-x_0}{x_1-x_0}\right)\left(\frac{x-x_2}{x_1-x_2}\right)\left(\frac{x-x_3}{x_1-x_3}\right) = \overset{x+1}{\left(\frac{x-(-1)}{0-(-1)}\right)}\overset{1-x}{\left(\frac{x-1}{0-1}\right)}\overset{\frac{2-x}{2}}{\left(\frac{x-2}{0-2}\right)} = \frac{1}{2}\left((x+1)(1-x)(2-x)\right) = -\frac{x^3}{3} + x^2 - \frac{x}{2} + 1$$

$$l_2 = \left(\frac{x-x_0}{x_2-x_0}\right)\left(\frac{x-x_1}{x_2-x_1}\right)\left(\frac{x-x_3}{x_2-x_3}\right) = \overset{\frac{x+1}{2}}{\left(\frac{x-(-1)}{1-(-1)}\right)}\overset{x}{\left(\frac{x-0}{1-0}\right)}\overset{2-x}{\left(\frac{x-2}{1-2}\right)} = \frac{1}{2}\overset{x^2-x^3+2x}{\left((x+1)\cdot x(2-x)\right)} = -\frac{x^3}{2} + \frac{x^2}{2} + x$$

$$l_3 = \left(\frac{x-x_0}{x_3-x_0}\right)\left(\frac{x-x_1}{x_3-x_1}\right)\left(\frac{x-x_2}{x_3-x_2}\right) = \overset{\frac{x+1}{3}}{\left(\frac{x-(-1)}{2-(-1)}\right)}\overset{\frac{x}{2}}{\left(\frac{x-0}{2-0}\right)}\overset{x-1}{\left(\frac{x-1}{2-1}\right)} = \frac{1}{6}\left((x+1)\,x\,(x-1)\right) = \frac{x^3}{6} - \frac{x}{6}$$

$$L(x) = \Sigma_{i=0}^{3} l_i y_i$$

$$= \frac{1}{81}\left(-\frac{x^3}{6} + \frac{x^2}{2} - \frac{x}{3}\right) + \frac{1}{3}\left(-\frac{x^3}{3} + x^2 - \frac{x}{2} + 1\right) + \left(-\frac{x^3}{2} + \frac{x^2}{2} + x\right) + \frac{1}{3}\left(\frac{x^3}{6} - \frac{x}{6}\right)$$

$$= -\frac{x^3}{486} + \frac{x^2}{162} - \frac{x}{243} - \frac{x^3}{9} + \frac{x^2}{3} - \frac{x}{6} + \frac{1}{3} - \frac{x^3}{2} + \frac{x^2}{2} + x + \frac{x^3}{18} - \frac{x}{18}$$

$$= x^3\left(-\frac{1}{486} - \frac{1}{9} - \frac{1}{2} + \frac{1}{18}\right) + x^2\left(\frac{1}{162} + \frac{1}{3} + \frac{1}{2}\right) - x\left(\frac{1}{243} + \frac{1}{6} - 1 + \frac{1}{18}\right) + \frac{1}{3}$$

$$= -\frac{271}{486}x^3 + \frac{68}{81}x^2 + \frac{188}{243}x + \frac{1}{3}$$

$$= \frac{68}{243}x^3 - \frac{14}{81}x^2 - \frac{188}{243} - \frac{1}{3}$$

b

$$x_i = \frac{1}{2}(a+b) + \frac{1}{2}(b-a)\tilde{x}_i$$

$$x_0 = \frac{1}{2}(-1+2) + \frac{1}{2}(2-(-1))\sqrt{\frac{3}{5} - \frac{2}{7}\sqrt{\frac{6}{5}}} = 1{,}304$$

$$x_1 = \frac{1}{2}(-1+2) + \frac{1}{2}(2-(-1))\left(-\sqrt{\frac{3}{5} - \frac{2}{7}\sqrt{\frac{6}{5}}}\right) = -0{,}304$$

$$x_2 = \frac{1}{2}(-1+2) + \frac{1}{2}(2-(-1))\sqrt{\frac{3}{5} + \frac{2}{7}\sqrt{\frac{6}{5}}} = 1{,}933$$

$$x_3 = \frac{1}{2}(-1+2) + \frac{1}{2}(2-(-1))\left(-\sqrt{\frac{3}{5} + \frac{2}{7}\sqrt{\frac{6}{5}}}\right) = -0{,}933$$